

# NARAMREDDY CHARAN KUMAR REDDY

charannaram1710@gmail.com | Portfolio: [naramcharan.me](https://naramcharan.me) | [github.com/NaramCharan](https://github.com/NaramCharan) | [linkedin.com/in/naramcharan](https://linkedin.com/in/naramcharan) | Gurugram, India

## PROFESSIONAL SUMMARY

3rd-year Computer Science student specializing in AI and Machine Learning, building end-to-end systems that reach production rather than stopping at a notebook metric. Experience spans medical-imaging deep learning, gradient-boosted forecasting across thousands of parallel series, and transformers built from scratch — plus the serving layer around them. Strong bias toward leakage-free validation and metrics that describe real behaviour.

## TECHNICAL SKILLS

|                                  |                                                                                                                                                                      |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Machine Learning:</b>         | XGBoost, LightGBM, Random Forest, Logistic Regression, Scikit-Learn, Optuna.                                                                                         |
| <b>Deep Learning &amp; CV:</b>   | PyTorch, Transfer Learning (ResNet / DenseNet / EfficientNet), CNNs, pydicom, Transformers (Self-Attention), BPE Tokenization.                                       |
| <b>Data Intelligence:</b>        | Pandas, NumPy, Feature Engineering, KNNImputer, Imbalance Handling, RobustScaler.                                                                                    |
| <b>Engineering &amp; Deploy:</b> | Python 3 (Advanced OOP), Data Structures, FastAPI, REST APIs, SQL (Schema Design), SQLAlchemy, Docker, Azure Container Apps, Git, Claude Code (agentic development). |

## ENGINEERING PROJECTS

**RSNA Pneumonia Detection | Chest X-Ray Screening | PyTorch, pydicom, FastAPI,** [NaramCharan/RSNA-Pneumonia-Detection](https://naramcharan.github.io/RSNA-Pneumonia-Detection)  
React, Docker, Azure

Live application: [rsna-app.salmonmeadow-7644e67e.eastasia.azurecontainerapps.io](https://rsna-app.salmonmeadow-7644e67e.eastasia.azurecontainerapps.io)

- Benchmarked DenseNet-121, EfficientNet-B2 and ResNet-34 under an identical two-stage transfer-learning protocol; the fine-tuned ResNet-34 reached **83% pneumonia recall, 0.79 F1 and 86% accuracy** on 1,836 held-out studies, against a 68% always-negative baseline.
- Diagnosed a RandomCrop(224) transform discarding 95% of every 1024×1024 radiograph while retaining the study-level label — removing it produced a larger gain than every architecture change in the project combined.
- Prevented patient-level leakage by splitting on unique patient IDs rather than rows, evaluating the test set once; deployed as one Docker container on Azure serving a React interface and REST API, cutting hosting cost 84%.

**Walmart Store Weekly Sales Forecasting | Python, Scikit-Learn, XGBoost,** [NaramCharan/Walmart-Store-Weekly-Sales-Forecasting](https://naramcharan.github.io/Walmart-Store-Weekly-Sales-Forecasting)  
LightGBM, Pandas

- Engineered a global multi-series forecasting pipeline predicting weekly sales across ~3,000 store-department combinations, achieving **95.55% validation R<sup>2</sup>** with an optimized LightGBM regressor.
- Built a custom recursive walk-forward inference system from scratch to project rolling and lag features over future validation horizons where actual sales are missing.
- Eliminated temporal data leakage by splitting before feature extraction and computing lag/rolling statistics strictly within each partition.

**E-Commerce Customer Churn Prediction | Python, Scikit-Learn, XGBoost,** [NaramCharan/ecommerce-customer-churn-prediction](https://naramcharan.github.io/ecommerce-customer-churn-prediction)  
PyTorch, Optuna

- Tuned tree classifiers with Optuna over 10-fold Stratified Cross-Validation, achieving **98.28% accuracy and 94.74% F1** on the final selected XGBoost production model; a 4-hidden-layer PyTorch network scored 98.76% and was rejected as not worth the compute.

**Handbuilt Transformer | Learning project | PyTorch, NLP** [NaramCharan/Handbuilt-Transformer](https://naramcharan.github.io/Handbuilt-Transformer)

- Built a mini GPT and a mini GPT tokenizer from scratch in PyTorch to learn how ChatGPT-style models work: a ~0.21M-parameter decoder-only Transformer (causal multi-head attention, 4 blocks) trained on Shakespeare, and a GPT-2-style byte-level BPE tokenizer with regex chunking, checked against **tiktoken**; ongoing — next, training on BPE tokens.

**Personal Portfolio — [naramcharan.me](https://naramcharan.me) | Next.js 16, React 19, TypeScript, Three.js,** [NaramCharan/NaramCharan.github.io](https://naramcharan.github.io)  
Tailwind v4, GitHub Actions

Live site: [naramcharan.me](https://naramcharan.me)

- Shipped a WebGL-hero portfolio with interactive project briefs, statically exported and auto-deployed to GitHub Pages; built by **directing Claude Code** across 75+ commits (architecture and product decisions mine, every change reviewed), lifting Lighthouse from 65 to 87.

## EDUCATION

**B.Tech in Computer Science Engineering (AI & Machine Learning) — 3rd Year** Graduation: May 2028  
GD Goenka University — Gurugram, India Current CGPA: 8.98 / 10.0

## CERTIFICATIONS

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|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| • Deep Learning Specialization – DeepLearning.AI (Jul 2026)              | • Claude Code: Agentic Coding Assistant – DeepLearning.AI, Anthropic (Jul 2026) |
| • Machine Learning Specialization – DeepLearning.AI, Stanford (May 2026) | • Databases and SQL for Data Science – IBM, Coursera (Nov 2025)                 |
| • Python for Everybody – Univ. of Michigan (Mar 2025)                    | • Prompt Engineering & GenAI – Google, Vanderbilt (Mar 2025)                    |